NLP
from
scratch
TEXT
UNDERSTAND
GENERATE
Understand Language
Generate Meaningful Responses

NATURAL LANGUAGE PROCESSING (NLP)

NLP is a field of AI that helps computers understand, interpret and generate human language.

GOAL:

To bridge the gap between Human Communication and Machine Understanding.

1) CORE NLP TASKS ☆

- Text Classification (e.g. Spam Detection)
- Sentiment Analysis (e.g. Positive / Negative)
- Named Entity Recognition (NER)
- Machine Translation
- Text Summarization
- Question Answering
- Language Generation
- Chatbots & Dialog Systems

2) NLP WORKFLOW ☆

1. Text Input
2. Preprocessing
3. Feature Representation
4. Model Training
5. Evaluation
6. Prediction / Output

3) TEXT PREPROCESSING STEPS ☆

- Lowercasing
- Tokenization
- Stopword Removal
- Stemming / Lemmatization
- Removing Punctuation & Special Characters
- Handling Numbers
- Spelling Correction

Text
↓
Clean Text

4) TOKENIZATION ☆

Breaking text into smaller units called tokens.

- Word Tokenization → "NLP is fun"
  → ["NLP", "is", "fun"]
- Sentence Tokenization → Split into sentences
- Subword Tokenization → e.g. using BPE, WordPiece

5) FEATURE REPRESENTATION ☆

- Bag of Words (BoW)
- TF-IDF (Term Frequency - Inverse Document Frequency)
- Word Embeddings (Word2Vec, GloVe)
- Contextual Embeddings (BERT, RoBERTa, etc.)

6) POPULAR NLP MODELS ☆

- Naive Bayes
- Logistic Regression
- SVM (Support Vector Machine)
- RNN (Recurrent Neural Network)
- LSTM / GRU
- Transformers (BERT, GPT, T5, etc.)

7) EVALUATION METRICS ☆

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- BLEU Score (for Translation)
- ROUGE Score (for Summarization)

8) NLP LIBRARIES IN PYTHON ☆

- NLTK (Natural Language Toolkit)
- spaCy
- TextBlob
- Gensim
- Hugging Face Transformers

9) APPLICATIONS OF NLP ☆

- Chatbots & Virtual Assistants
- Language Translation
- Sentiment Analysis
- Text Summarization
- Voice Recognition
- Spam Detection
- Search Engine
- Content Recommendation

10) CHALLENGES IN NLP ☆

- Ambiguity in Language
- Sarcasm & Irony
- Context Understanding
- Multilingual Support
- Data Quality
- Bias in Models

NLP is powering the future of communication between humans and machines. The better we understand language, the smarter the machines become.

NATURAL LANGUAGE PROCESSING (NLP)

11) PARTS OF SPEECH (POS) TAGGING ☆

Assigns a part of speech to each word in a sentence.

Common Tags:

- Noun (NN)
- Verb (VB)
- Adjective (JJ)
- Adverb (RB)
- Pronoun (PRP)
- Preposition (IN)
- Conjunction (CC)
- Determiner (DT)
- Interjection (UH)

Noun (NN)
Verb (VB)
Adj. (JJ)
Adv. (RB)

12) NAMED ENTITY RECOGNITION ☆

Identifies and classifies important entities in text into predefined categories.

Common Entities:

- Person (PER)
- Organization (ORG)
- Location (LOC)
- Date / Time (DATE)
- Money (MONEY)
- Percent (PERCENT)
- Product (PRODUCT)

Example:
"Elon Musk" founded SpaceX in California on May 30, 2020."

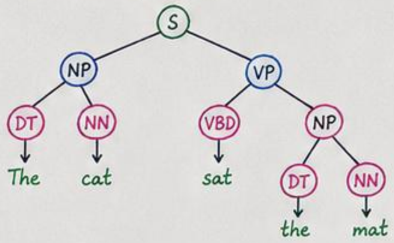
PER ORG LOC DATE

13) SYNTACTIC ANALYSIS ☆

Analyzes the grammatical structure of a sentence.

Techniques:

- Parsing
- Constituency Parsing
- Dependency Parsing



14) SEMANTIC ANALYSIS ☆

Focuses on the meaning of the text.

Tasks:

- Word Sense Disambiguation (WSD)
- Semantic Role Labeling (SRL)
- Coreference Resolution
- Sentiment / Intent Understanding

15) WORD EMBEDDINGS ☆

Convert words into dense vector representations capturing meaning.

Popular Methods:

- Word2Vec (CBOW, Skip-gram)
- GloVe
- FastText
- BERT Embeddings

king → 0.21, -0.74, 0.63, ⋮
queen → 0.11

16) LANGUAGE MODELS ☆

Models that predict the next word in a sequence.

Types:

- N-gram Models
- Statistical Language Models
- Neural Language Models
- Pretrained LMs (GPT, BERT, LLaMA, etc.)

Next word?

17) TEXT CLASSIFICATION ☆

Assigns a category/label to a given text.

Examples:

- Spam vs Not Spam
- Sentiment (Positive / Negative / Neutral)
- Topic Classification

SPAM → Spam, Ham, Promotions

18) SEQUENCE LABELING TASKS ☆

Assigns a label to each word in a sequence.

Examples:

- POS Tagging
- NER
- Chunking

John	lives	in	New York	City	.
↓	↓	↓	↓	↓	
PER	VBZ	IN	LOC	LOC	.

19) MACHINE TRANSLATION ☆

Automatically translates text from one language to another.

Approaches:

- Rule-based MT
- Statistical MT (SMT)
- Neural MT (NMT – Seq2Seq, Transformer)

A 文

20) SENTIMENT ANALYSIS ☆

Determines the emotional tone behind the text.

Classes:

Positive Neutral Negative

21) TOPIC MODELING ☆

Discovers hidden topics in a collection of documents.

Popular Algorithms:

- LDA (Latent Dirichlet Allocation)
- NMF (Non-negative Matrix Factorization)

Topic 1
Topic 2
⋮
Topic K

22) CHATBOTS & DIALOG SYSTEMS ☆

Build conversational agents that can understand and respond to users.

Components:

- Intent Detection
- Entity Extraction
- Dialogue Management
- Response Generation

Hello!

☆ Understanding language is the first step towards building intelligent machines.

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23) LANGUAGE MODELING ☆

Predicts the probability of a sequence of words.

Types:

- Unigram Model
- Bigram Model
- Trigram Model
- N-gram Model

The cat sat on the mat

P(next word | previous words)

24) PERPLEXITY ☆

Measures how well a probability model predicts a sample.

$$\text{Perplexity} = 2^{-\frac{1}{N}\sum_{i=1}^{N} \log_2 P(w_i)}$$

Lower perplexity → Better model

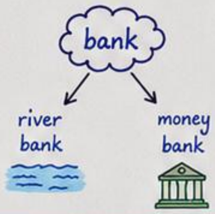
Higher perplexity → Poorer model

25) WORD SENSE DISAMBIGUATION (WSD) ☆

Finds the correct meaning of a word based on its context.

Approaches:

- Knowledge-based
- Supervised Learning
- Embedding-based



26) COREFERENCE RESOLUTION ☆

Identifies when different expressions refer to the same entity.

Example:

Riya went to the market. She bought some fruits.

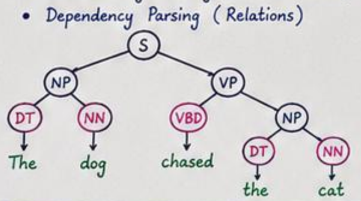
Same Entity

27) PARSING ☆

Analyzes the grammatical structure of a sentence.

Types:

- Constituency Parsing (Tree structure)
- Dependency Parsing (Relations)

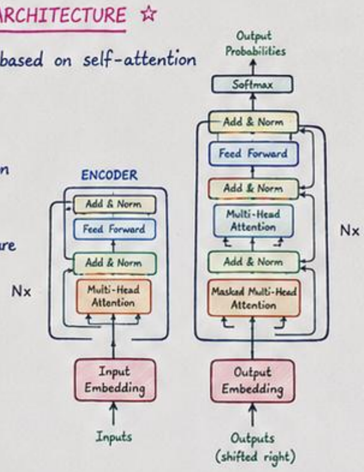


28) TRANSFORMERS ARCHITECTURE ☆

A deep learning model based on self-attention mechanism.

Key Components:

- Multi-Head Self-Attention
- Positional Encoding
- Encoder – Decoder Structure
- Feed Forward Networks



29) ATTENTION MECHANISM ☆

Allows the model to focus on relevant parts of the input.

Types:

- Scaled Dot-Product Attention
- Multi-Head Attention

30) Positional Encoding ☆

Adds position information to word embeddings since transformers have no recurrence.

Formula:

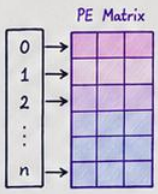
$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

Where,

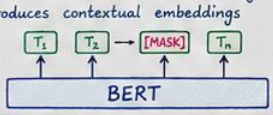
pos = position, i = dimension index,

d_{model} = embedding dimension



31) BERT (Bidirectional Encoder Representations from Transformers) ☆

- Pre-trained using Masked Language Modeling (MLM) and Next Sentence Prediction (NSP)
- Bidirectional context understanding
- Produces contextual embeddings



32) GPT (Generative Pre-trained Transformer) ☆

- Decoder-only architecture
- Trained using causal (autoregressive) language modeling
- Generates text one token at a time

$$T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow \cdots \rightarrow T_n$$

33) TEXT GENERATION TECHNIQUES ☆

- Greedy Search → picks highest probability token
- Beam Search → keeps top k candidates
- Top-k Sampling → sample from top k tokens
- Top-p (Nucleus) Sampling → sample from top tokens with cumulative prob p
- Temperature Sampling → controls randomness

34) FINE-TUNING & TRANSFER LEARNING ☆

- Fine-tuning adapts a pre-trained model to a specific task with smaller dataset.
- Transfer learning saves time, data and improves performance.

☆ NLP combines linguistics, statistics and deep learning to help machines understand human language.

NATURAL LANGUAGE PROCESSING (NLP)

35) LANGUAGE UNDERSTANDING TASKS ☆

Help machines understand the meaning behind text.

Examples:

- Intent Classification
- Question Classification
- Natural Language Inference (NLI)
- Textual Entailment
- Semantic Similarity

What is NLP?
↓
Intent: Information Request

36) SLOT FILLING ☆

Extracts specific information (data) from the user input based on predefined slots.

Example:

User Input	Extracted Slots
Book a flight from Hyderabad to Delhi on 25 May.	From : Hyderabad To : Delhi Date : 25 May

37) DIALOGUE STATE TRACKING ☆

Tracks the context and state of a conversation over multiple turns.

User: I want to book a hotel. → State Tracker → System: Sure! Where do you want to go?

State:
intent = book_hotel
slot(city) = ?
slot(date) = ?

38) TEXT NORMALIZATION ☆

Converts informal or noisy text into a standard format.

Examples:

- "gud mrng" → "good morning"
- "12th Jan'24" → "12 January 2024"
- "can't" → "cannot"

39) LEMMATIZATION VS STEMMING ☆

Stemming	Lemmatization
• Removes word endings • May not be real words • Simple & Fast	• Converts to root word (dictionary form) • Always real words • Slower but accurate

Examples:

Word	Stemming	Lemmatization
running	run	run
better	better	good
studies	studi	study

40) STOP WORDS ☆

Common words in a language that do not carry much meaning and are often removed.

Examples:

the, is, in, on, at, a, an, and, or, but, to, of etc.

the is in on at and

41) N-GRAMS ☆

A sequence of N items (words/characters) from a given text.

Types:

- Unigram (1-gram): one word
  e.g., I love NLP
- Bigram (2-gram): two words
  e.g., I love, love NLP
- Trigram (3-gram): three words
  e.g., I love NLP

Text: I love NLP → 1-gram 2-gram 3-gram

42) HANDLING OUT-OF-VOCABULARY (OOV) ☆

Words that are not present in the vocabulary.

Solutions:

- Use <UNK> token
- Subword Tokenization (BPE, WordPiece)
- Character-level models

Word: unbelievable
↓
<UNK>

43) SUBWORD TOKENIZATION ☆

Breaks words into smaller units (subwords) to handle rare or unseen words.

Methods:

- Byte Pair Encoding (BPE)
- WordPiece
- SentencePiece

Example (BPE):

unbelievable → un + belief + able

44) LANGUAGE IDENTIFICATION ☆

Detects the language of a given text.

Example:

"Hola, ¿Cómo estás?"
↓
Detected Language: Spanish

Hello

नमस्ते

45) CODE-MIXING & CODE-SWITCHING ☆

Mixing multiple languages in a single sentence or switching between languages.

Examples:

- Code-Mixing: "Kal meeting hai, ready ho na?" (Hindi + English)
- Code-Switching: "I will call you later, ठीक है?" (Switching between English & Hindi)

Hello!

कैसे हो?

46) PRAGMATICS IN NLP ☆

Understanding language in context, considering speaker's intent, tone, and real-world knowledge.

Example:

- "Can you pass the salt?"
  → Not a question, but a request.

Can you pass the salt?

NLP is not just about processing text, it's about understanding humans.
The more context we give machines, the better they understand us. ♥

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47) COREFERENCE RESOLUTION ☆

Identifies when different expressions refer to the same entity in a text.

Example:

Arjun went to the store. He bought milk.

Arjun, He → Same Entity

Approaches:

- Mention Pair Models
- Entity Centric Models
- Neural Coreference Models

48) DEPENDENCY PARSING ☆

Shows grammatical structure by connecting words with dependency relationships.

Example:

I love NLP .

PRP VBP NNP .

nsubj: love → I; dobj: love → NLP; punct: NLP → .

Common Relations:

nsubj (nominal subject), dobj (direct object), amod (adjectival modifier), advmod, etc.

49) SEMANTIC ROLE LABELING (SRL) ☆

Identifies predicate (verb) and its roles like who did what to whom.

Example:

Riya gave a book to Meera .

Riya → ARG0 (giver); gave → V (verb); a book → ARG1 (thing); Meera → ARG2 (recipient)

Applications:

- Information Extraction
- Question Answering
- Text Summarization

50) INFORMATION EXTRACTION (IE) ☆

Extracts structured information from unstructured or semi-structured text.

Types:

- Entity Extraction
- Relation Extraction
- Event Extraction

Example:

Apple Inc. was founded in Cupertino by Steve Jobs in 1976.

ORG | LOCATION | PERSON | DATE

51) QUESTION ANSWERING (QA) ☆

Automatically answers questions based on the given context.

Types:

- Extractive QA – answer in the context
- Abstractive QA – generate answer

Example:

Context: The Eiffel Tower is in Paris. It was completed in 1889.

Q: Where is the Eiffel Tower?

A: Paris

52) TEXT SUMMARIZATION ☆

Reduces long text into shorter version while keeping important information.

Types:

- Extractive Summarization
- Abstractive Summarization

Example:

Original Text

NLP is a field of AI that enables computers to understand, interpret and generate human language. It combines linguistics, machine learning and deep learning techniques.

Summary

NLP enables computers to understand, interpret and generate human language using AI.

53) SENTENCE EMBEDDINGS ☆

Converts sentences into dense vectors capturing semantic meaning.

Popular Models:

- Sentence-BERT (SBERT)
- Universal Sentence Encoder (USE) →
- InferSent
- LaBSE

Embedding Vector

0.13
-0.45
0.87
0.21
⋮
-0.34

54) RETRIEVAL AUGMENTED GENERATION (RAG) ☆

Combines retrieval and generation to produce more accurate and context-aware responses.

Pipeline:

User Query → Retrieve Relevant Documents → Augment Prompt → LLM Generate Answer → Response

Benefits:

- Reduces hallucinations
- Uses up-to-date information
- Improves factual accuracy

55) NLP EVALUATION CHALLENGES ☆

- Ambiguity & Context Dependence
- Sarcasm & Irony
- Low-resource Languages
- Bias & Fairness
- Hallucination in Generative Models
- Domain Adaptation

Language is dynamic, contextual and deeply human. NLP aims to bridge the gap between humans and machines. ♥

From understanding words to understanding humans – NLP continues to evolve and empower machines to communicate like us. ♥

Page 6 (Advanced)

MULTIMODAL NLP (M-NLP)

Multimodal NLP enables models to understand and generate responses using multiple types of data (modalities) like text, images, audio, and video together.

1) WHAT IS MULTIMODAL NLP? ☆

Traditional NLP works only with text. Multimodal NLP combines two or more modalities to understand context better and generate more accurate and human-like responses.

Text | Image | Audio | Video | More

2) COMMON MODALITIES ☆

- Text – words, sentences, documents
- Image – photos, diagrams, charts
- Audio – speech, sounds
- Video – frames, actions, scenes
- Sensor Data – GPS, IoT, signals, etc.

Combining these helps models understand richer context!

3) WHY MULTIMODAL NLP? ☆

- Closer to human communication
- Better understanding of context
- Handles real-world complex data
- Improves accuracy & robustness
- Enables new applications

More Context = Better Understanding

4) MULTIMODAL NLP ARCHITECTURE ☆

INPUT MODALITIES: Text, Image, Audio, Video, ⋮

→ **MODALITY ENCODERS**:
- Text Encoder (BERT, RoBERTa)
- Image Encoder (ViT, ResNet)
- Audio Encoder (Wav2Vec, Whisper)
- Video Encoder (TimeSformer, ViViT)

→ **ALIGNMENT / FUSION LAYER**: Cross-modal Attention / Fusion

→ **MULTIMODAL REPRESENTATION**

→ **LLM / DECODER** (GPT, LLaMA, PaLM, etc.)

→ **OUTPUT (RESPONSE)**: Here is the answer based on the image and question.

Encoders convert each modality → Fusion layer combines → LLM generates final response

5) EXAMPLES OF MULTIMODAL NLP ☆

- Image + Text → Visual Question Answering (e.g., What is in this picture?)
- Audio + Text → Speech to Text + Understanding (e.g., Transcribe and summarize)
- Video + Text → Video Captioning / Q&A (e.g., Describe this video)
- Image + Text + Audio → Multimodal Chatbots / Assistants

Used in: Chatbots, Content Analysis, Accessibility, Healthcare, Education, Search, Autonomous Systems & more.

6) POPULAR MULTIMODAL MODELS ☆

- CLIP (Text + Image)
- BLIP / BLIP-2 (Image-Text Understanding & Generation)
- Flamingo (DeepMind)
- LLaVA (Large Language and Vision Assistant)
- PaLI (Google)
- GPT-4V / Gemini (Vision + Text)
- Whisper (Audio + Text)
- Video-LLaVA (Video + Text)

7) CHALLENGES ☆

- Data alignment across modalities
- High computational cost
- Handling missing or noisy modalities
- Bias & fairness in multimodal data
- Long context understanding

8) FUTURE OF MULTIMODAL NLP ☆

- More human-like AI interactions
- Real-time multimodal assistants
- Better accessibility & inclusivity
- Strong impact on Robotics, AR/VR, Healthcare, Education & beyond!

Multimodal NLP is the bridge between how machines perceive and how humans communicate! ♥

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